1.2 Excess carriers – non-equilibrium situation

So far we have discussed the case of the semiconductor in thermal equilibrium, where the product of electron and hole concentrations is a constant for a particular material at a given temperature. However, if excess carriers are introduced in a semiconductor, so that $np > n_i^2$, we have a non-equilibrium situation. Excess carriers can be introduced in a semiconductor by shining light (optical excitation) or forward-biasing a p-n junction (to be discussed in detail in Chapter4). The process of introducing excess carriers is called injection. In case of optical excitation, if the photons have energy $h\nu > E_g$, they are absorbed in the semiconductor, resulting in excitation of electrons from the valence band to the conduction band. Since the optical excitation results in additional EHPs being generated, a new steady state is reached where the recombination rate is equal to the total generation rate (thermal + optical). The electron and hole concentrations in this steady state are more than their thermal equilibrium values. These additional carriers are called **excess carriers**. The magnitude of the excess carriers relative to the equilibrium majority carrier concentration determines the level of injection. For example, let us consider an n-type silicon sample with $N_D = 10^{15}/cm^3$ at 300K. The majority carrier concentration at thermal equilibrium is $n_{no} = 10^{15}/cm^3$ assuming complete ionization and the minority carrier concentration is $p_{no} = 2.25 \times 10^5/cm^3$. (The subscript 'n' indicates that we are referring to an n-type semiconductor and 'o' refers to the thermal equilibrium condition). Now let us suppose that by optical excitation $10^{12}/cm^3$ excess minority carriers are injected in this sample. The excess electron concentration n_n' must be equal to the excess hole concentration p_n' since excess holes and electrons are generated in pairs. Thus while the minority carrier concentration in this sample has increased by

nearly seven orders of magnitude (from $2.25 \times 10^5/cm^3$ to $10^{12}/cm^3$), the increase in the electron concentration is negligible. This condition where the excess carrier concentration is small compared to the majority carrier concentration i.e. $n_n' = p_n' << n_{no}$ is referred to as **low level injection**. The other case, where $n_n' = p_n' \approx n_{no}$ or greater than n_{no} is called **high level injection** and is more difficult to analyze. We shall restrict our discussion mostly to low level injection cases in this book.

1.4.1 Quasi Fermi Level or IMREF

When excess carriers are generated, since $np \neq n_i^2$, eqns. (1.27) and (1.28) are no longer valid and the concept of a Fermi energy level becomes meaningless. However it is desirable to retain the same form of expressions for the electron and hole concentrations. Therefore we can define separate **quasi-Fermi levels** (E_{Fn} and E_{Fp}, also referred to as IMREFs, that is Fermi spelt backwards) for electrons and holes for the non-equilibrium cases. Writing the expressions for the electron and hole concentrations for the non-equilibrium case in steady state, in the same form as in eqns. (1.27) and (1.28), we have

$$n = n_i \exp\left(\frac{E_{Fn} - E_i}{kT}\right) \ldots\ldots\ldots (1.49)$$

$$p = n_i \exp\left(\frac{E_i - E_{Fp}}{kT}\right) \ldots\ldots\ldots (1.50)$$

From eqns.(1.49) and (1.50), we see that

$$n.p = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right) \ldots\ldots\ldots (1.51)$$

Therefore, if the quasi Fermi level for electrons is above that for the holes, $n.p > n_i^2$ and we have excess carrier injection. If $n.p = n_i^2$, we have the equilibrium situation with a

single Fermi level. If $n.p < n_i^2$, E_{Fn} is below E_{Fp} and instead of carrier injection, we have carrier extraction. More about this will be discussed in Chapter 4.

1.4.2 Generation and Recombination of carriers and the concept of lifetime

Whenever the thermal equilibrium condition is disturbed, processes exist to restore the equilibrium condition. In the case of excess carrier injection, the mechanism that restores equilibrium is **recombination** of the injected minority carriers with majority carriers. Recombination can be *direct* (also called band-to-band) which involves a transition of carrier between E_c and E_v or *indirect* involving an intermediate trap level (called the recombination centre) in the bandgap. Generally direct recombination is the dominant process in a direct bandgap semiconductor (e.g. GaAs, InP) while indirect recombination is prevalent where the material has an indirect band gap (e.g. Si, Ge). The energy released from the recombination process can be emitted as a photon (radiative recombination) or as a phonon, i.e. dissipated as heat to the lattice (non-radiative recombination). Let us consider direct recombination first. In a direct bandgap semiconductor, the continuous thermal vibration of lattice atoms causes some bonds between neighbouring atoms to break. EHPs are thus generated, i.e. thermal energy enables an electron to make an upward transition from the valence band to the conduction band leaving a hole in the valence band. This is called **generation of carriers and is represented by a generation rate G_{th}** (i.e. number of EHPs generated per unit volume per second). On the other hand, when an electron makes a transition from E_C down to E_V, an EHP is annihilated and this is called recombination of carriers. Let **R_{th} represent the recombination rate at thermal equilibrium**. As mentioned earlier, R_{th} is proportional to

the electron concentration in the conduction band and the hole concentration in the valence band. Therefore, for an n-type semiconductor we can write

$$R_{th} = \beta n_{no}\, p_{no} \text{……..(1.52)}$$

where β is the constant of proportionality and n_{no} and p_{no} denote the electron and hole concentrations respectively under thermal equilibrium. At thermal equilibrium, $G_{th} = R_{th}$ so as to keep the pn product constant. Therefore

$$G_{th} = R_{th} = \beta n_{no} p_{no} \text{……..(1.53)}$$

When we shine light on this semiconductor to generate EHPs at a rate of G_L per cm^3-sec, the carrier concentrations rise above their thermal equilibrium value to n_n and p_n. Then the new recombination rate is given by

$$R = \beta n_n p_n = \beta (n_{no} + n_n')(p_{no} + p_n') \text{……..(1.54)}$$

where n_n' and p_n' are the excess electron and hole concentrations. The new generation rate is given by

$$G = G_{th} + G_L \text{……(1.55)}$$

The net rate of change in the minority carrier concentration (holes for n-type material) is the difference of the generation and recombination rates, and is given by

$$\frac{dp_n}{dt} = G - R = G_{th} + G_L - R \text{ ……(1.56)}$$

Now at steady state, $\frac{dp_n}{dt} = 0$. Therefore from eqn. (1.56), we can write

$$G_L = R - G_{th} = U \text{……(1.57)}$$

where U is the **excess recombination rate**. Substituting the values of G_{th} and R from eqns. (1.53) and (1.54) respectively into eqn. (1.57), we have

$$U = \beta (n_{no} + n_n')(p_{no} + p_n') - \beta (n_{no} p_{no}) = \beta (n_{no} + p_n' + p_{no})\, p_n' \text{ …..(1.58)}$$

since $n_n' = p_n'$ as holes and electrons are generated in pairs. Now, for the low injection case in an n-type semiconductor, $p_n' << n_{no}$ and $p_{no} << n_{no}$. Therefore the net recombination rate can be approximated as

$$U = \beta n_{no} p_n' = \beta n_{no}(p_n - p_{no}) = \frac{p_n - p_{no}}{\tau_p} = \frac{\text{excess carrier concentration}}{\text{lifetime}} \quad \ldots\ldots\ldots(1.59)$$

where τ_p is defined as the **lifetime** of the minority carriers and is a constant for the particular semiconductor sample. Since U represents the excess recombination rate over and above the thermal equilibrium value, we see from eqn.(1.59) that U = 0 when $p_n = p_{no}$, i.e. the minority concentration is equal to the thermal equilibrium value. In eqn.(1.59), we have considered excess carriers. However, if carriers are extracted, so that $p_n < p_{no}$, U is negative. This implies that the thermal generation rate is more than the recombination rate, and –U represents the **excess generation rate**.

In order to understand the concept of lifetime, let us consider an n-type semiconductor illuminated by light, which results in additional EHPs being generated throughout the sample at a constant rate of G_L per unit volume per unit time. In the steady state, from eqns. (1.57) and (1.59), $G_L = U = (p_n - p_{no})/\tau_p$. Therefore

$$p_n = p_{no} + \tau_p G_L \ldots\ldots(1.60)$$

We see that the excess carrier concentration is proportional to the lifetime. If we consider the majority carriers, we can write $n_n = n_{no} + n_n'$. In the low injection case, since $n_n' = p_n' < n_{no}$, we have $n_n \approx n_{no}$. In other words, **the minority carrier concentration changes significantly while majority carrier concentration is almost unaffected**.

At an arbitrary time instant (t = 0), let the light be switched off. Now, since $G_L = 0$, from eqns. (1.56), (1.57) and (1.59) we can write that for t > 0,

$$\frac{dp_n}{dt} = G_{th} - R = -U = -\frac{p_n - p_{no}}{\tau_p} \quad \ldots\ldots(1.61)$$

The solution of the above equation is of the form

$$p_n - p_{no} = C\exp\left[-\left(\frac{t}{\tau_p}\right)\right] \quad \ldots\ldots\ldots(1.62)$$

where C is the integration constant. In order to evaluate C, we note that at t = 0, $(p_n - p_{no})$ $= \tau_p G_L$. Therefore, we can write

$$p_n = p_{no} + \tau_p G_L \exp\left[-\left(\frac{t}{\tau_p}\right)\right] \quad \ldots\ldots\ldots(1.63)$$

So we see that once the light is switched off, with time, the light-generated minority carriers recombine with the majority carriers to bring the carrier concentrations back to their thermal equilibrium values. This decay of excess carriers occurs exponentially with a **time constant τ_p, defined as the lifetime of the minority carriers**. If the excess carriers have a longer lifetime, it takes more time for the carrier concentration to revert to the thermal equilibrium value. On the other hand, if the excess carriers have a shorter lifetime, they 'die' quickly, and thermal equilibrium values are attained faster. Please note that although in a direct recombination process, the majority and minority excess carriers decay at the same rate, i.e. $\tau_n = \tau_p$, the pace of recombination is set by the minority carriers as they are in short supply. Thus the recombination process is ultimately characterised by the minority carrier lifetime. Fig. 1.17 shows the variation of the majority and minority carrier concentrations as a function of time.

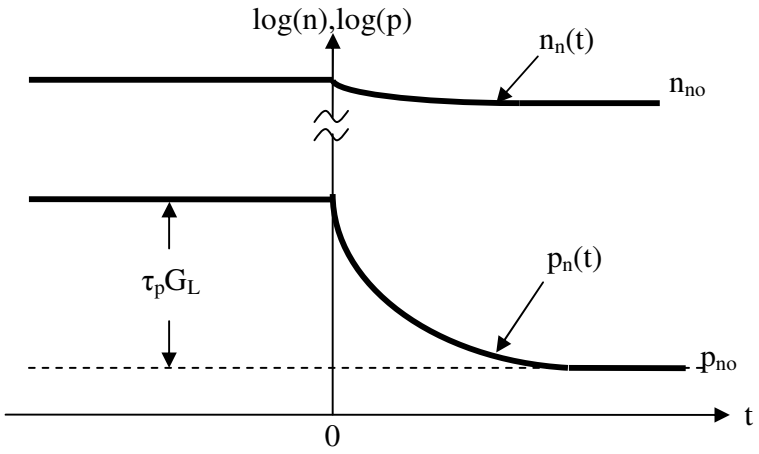


Fig. 1.17 Variation of electron and hole concentrations with time

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Q 1.4 Why is the constant τ called the lifetime?

τ is called the lifetime as it can be considered to be the average time between the generation and recombination of a carrier. We shall explain this with an analogous example. Let us imagine the existence of a village with 1000 inhabitants where each inhabitant lives for exactly 50 years. In this village, the birth rate was strictly controlled and was equal to the death rate, so that the population was fixed at 1000. In a particular year, say 1900, the villagers decided to allow 5 additional babies to be born every year. So for the next 50 years, 250 additional babies were born, raising the total population of the village to 1000 + 50 x 5 = 1250. But after 50 years elapsed, the death rate suddenly went up by 5 as the 5 additional babies born 1900 onwards started dying from 1950. So again the birth rate became equal to the death rate and a steady state was reached at a population of 1250. The analogy to the optical generation case in semiconductor is quite obvious. We could obtain the final steady state population in the village using eqn.(1.60),

where $p_n = 1000$, $G_L = 5$ and $\tau = 50$. This also brings out why τ is referred to as the lifetime.

The concept of lifetime can also be explained with the help of eqn.(1.59), where the rate of recombination is given as the excess carrier concentration divided by lifetime. This clearly implies that the lifetime is the average time before a carrier recombines.

Example 1.6: Consider an n-type silicon sample with a doping concentration $N_D = 10^{16}/cm^3$. If it is illuminated such that $10^{18}/cm^3/sec$ EHPs are generated, find out the majority and minority carrier concentrations at steady state. Assuming $\tau_p = 10^{-6}$ sec, plot the decay of the excess carriers with time once the light source is switched off.

Assuming complete ionisation, at thermal equilibrium the electron concentration $n_{no} = N_D = 10^{16}/cm^3$. Therefore, the hole concentration $p_{no} = n_i^2/n_{no} = 2.25x10^4/cm^3$. Now, when the sample is illuminated such that $G_L = 10^{18}/cm^3/sec$, in steady state, from eqn.(1.60), we have $p_n' = n_n' = \tau_p G_L = 10^{12}/cm^3$. Therefore $p_n = p_{no} + p_n' \approx p_n' = 10^{12}/cm^3$ and $n_n = n_{no} + n_n' \approx n_{no} = 10^{16}/cm^3$. Please note that there is a large change in the minority carrier concentration while the majority carrier concentration remains virtually unaffected.

Now when the light is switched off, excess carriers will decay with a time constant τ_p as shown in the Fig.1.17. The minority carrier concentration will decay exponentially from $10^{12}/cm^3$ to the thermal equilibrium value of $2.25x10^4/cm^3$. However, the majority carrier concentration remains almost constant throughout at about $10^{16}/cm^3$.

Example 1.7: For the n-type silicon sample in Example 1.6, show the Fermi level positions before and after the light source is switched on at room temperature. (Given: kT = 0.026eV and $n_i = 1.5x10^{10}/cm^3$ at room temperature)

Before light is switched on, the sample is in thermal equilibrium and the Fermi level position is given by eqn.(1.29). Assuming complete ionization, we have $n \approx N_D = 10^{16}/cm^3$. Therefore, from eqn.(1.29) at room temperature

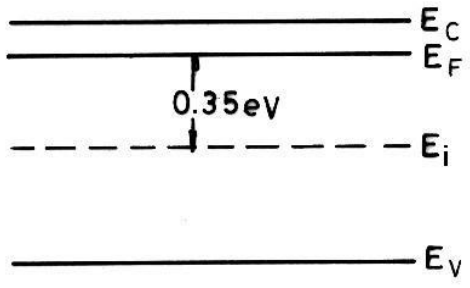
$$E_F = E_i + 0.026\ln\left(\frac{10^{16}}{1.5x10^{10}}\right) = E_i + 0.35eV$$

When light is switched on, from Example 1.6, we have $n_n = 10^{16}/cm^3$ and $p_n = 10^{12}/cm^3$. Since the sample is no longer in thermal equilibrium ($n.p \neq n_i^2$), the Fermi level splits into the Electron and Hole Quasi Fermi levels, whose positions are given by eqns. (1.49) and (1.50). From these equations, we have

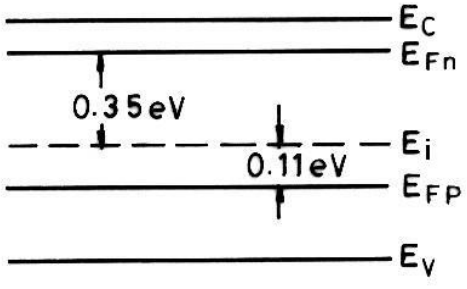
$$E_{Fn} = E_i + kT\ln\left(\frac{n}{n_i}\right) = E_i + 0.026\ln\left(\frac{10^{16}}{1.5x10^{10}}\right) = E_i + 0.35eV$$

$$E_{Fp} = E_i - kT\ln\left(\frac{p}{n_i}\right) = E_i - 0.026\ln\left(\frac{10^{12}}{1.5x10^{10}}\right) = E_i - 0.11eV$$

We find that the position of E_{Fn} coincides with E_F, since the majority carrier concentration is almost unchanged. Also, $(E_{Fn} - E_{Fp})$ is positive showing the presence of excess carriers. The positions of the Fermi level and the Quasi Fermi levels are shown below.



Before illumination



After illumination

1.4.3 Indirect recombination

For an indirect bandgap semiconductor, the process of direct recombination with an associated photon emission is unlikely, since, when the electron makes a transition from E_c to E_v, not only the energy but also the momentum needs to be conserved. Photons, being light particles (literally!), cannot assist in momentum conservation. In such cases, the dominant recombination process is indirect transition via localized energy states between E_C and E_V. As the lifetimes in indirect semiconductors are generally high, these states, which act as steps, are sometimes deliberately created by introducing special impurities (Au, Pt in silicon as well as N in GaAsP). As the transition probability depends on the energy difference between the step and E_C (or E_V), they can substantially increase the recombination rate and consequently reduce the lifetime.

1.4.4 Surface recombination

In Sections 1.4.2 and 1.4.3, we have essentially discussed about the recombination process in the bulk of a semiconductor. A similar process also occurs at the semiconductor surface. The discontinuity of the lattice structure at the semiconductor surface introduces a large number of energy states in the forbidden gap, called **surface states**. The presence of these states greatly enhances the recombination rate at the surface. In addition, there may be adsorbed ions, molecules or damages in the surface layer, which also increase the recombination rate. The recombination rate at the surface per unit area (U_s) can be expressed in a similar form to eqn.(1.59) as

$$U_s = S\,[p_n - p_{no}] \ldots\ldots\ldots (1.64)$$

Since U_s and p_n have the dimensions of $cm^{-2}s^{-1}$ and cm^{-3}, the dimension of the constant S is cm/s. Since S has the same dimension as velocity, it is called the **surface**

recombination velocity and it plays a similar role as lifetime in bulk recombination. Of course, a higher value of S indicates a higher recombination rate.

1.5 Mobility of carriers

1.5.1 Effect of electric field on carrier movement

In a semiconductor, the charge carriers (i.e. electrons and holes) are in constant motion even at thermal equilibrium. The thermal motion of an electron at room temperature may be visualized as random scattering from lattice atoms, impurities, other electrons and defects. This random movement of an electron leads to a net zero displacement over a sufficiently long period of time. Schematically, this can be shown as in Fig. 1.18(a), where the trajectory of an electron consists of a series of straight lines between collisions. The average time between collisions is called the **mean free time** and the average distance travelled between collisions is the **mean free path**. The root-mean-square (rms) value of the random **thermal velocity** (v_{th}) is of the order of 10^7cm/sec for most semiconductors.

When an electric field (ε) is applied, the situation changes. Each electron now experiences a force ($-q\varepsilon$) from the field and is consequently accelerated between each collision in a direction opposite to the field, showing a net displacement of the electron. This situation is schematically represented in Fig. 1.18(b). Thus, we see that an additional velocity component is superposed on the random thermal velocity. This additional component is called the **drift velocity ($\mathbf{v_d}$)**, which is defined as the net displacement per unit time.

Thus, in presence of an electric field, an electron acquires a drift velocity and its resultant velocity is therefore the vector sum of v_{th} and v_d. Since the mean free path does

not alter due to the application of field, this increase in electron velocity reduces the mean free time. Consequently, the electron suffers more frequent collisions and therefore loses more energy to the lattice. This acts like a resistive force, which does not allow a continuous increase of the carrier velocity, but rather, in equilibrium an electron acquires an average drift velocity. Assuming that in steady state, all momentum gained between collisions is lost to the lattice, we can write

$$m_e^* v_d = -q\varepsilon\tau_c \text{.....(1.65)}$$

where m_e^* is the effective mass of the electron and τ_c is the mean free time between collision. Therefore

$$v_d = -\frac{q\tau_c}{m_e^*}\varepsilon = -\mu_n\varepsilon \text{ .....(1.66)}$$

Thus we see that the average drift velocity of the electrons is proportional to the applied electric field and this constant of proportionality (μ_n) is called the **mobility of electrons**. The negative sign in the above equation indicates that the direction of v_d is opposite to that of the applied field. Analogously, for holes we can write

$$v_d = \frac{q\tau_c}{m_h^*}\varepsilon = \mu_p\varepsilon \text{ .....(1.67)}$$

where m_h^* is the effective mass of holes. As holes carry positive charge, their movement will be in the same direction as the electric field and hence there is no negative sign. Usually $m_e^* << m_h^*$, and therefore the mobility of electrons is larger than that of holes for any semiconductor. Typical values of electron and hole mobilities at 300K for some common semiconductors are listed in Table 1.4.

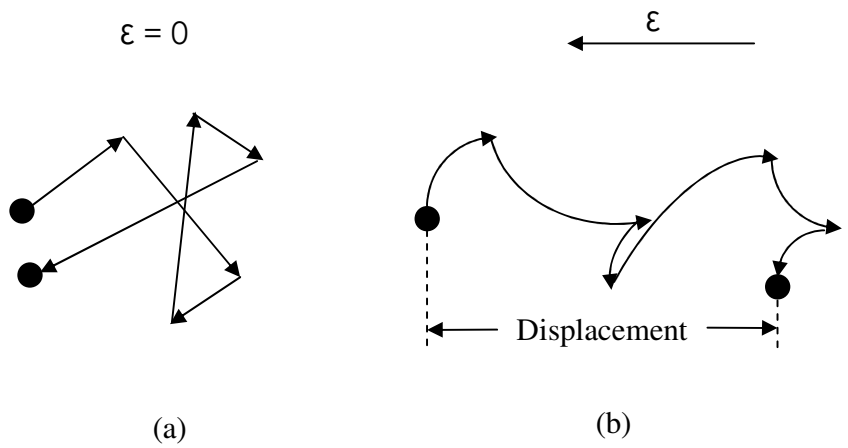


Fig. 1.18 Possible trajectory of an electron (a) in the absence of electric field and (b) in the presence of electric field

Semiconductor	**Electron mobility (cm^2/V-sec)**	**Hole mobility (cm^2/V-sec)**
Silicon	1450	500
Germanium	3900	1900
Gallium Arsenide	8500	400
Indium Phosphide	4600	150
Indium Arsenide	33000	400

Table 1.4 Electron and hole mobilities in some common semiconductors

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Q 1.5 While calculating effective density of states of silicon in Section 1.3, m_e^* has been considered greater than m_h^*, so that N_C is greater than N_V. On the other hand, while calculating mobilities of electrons and holes, m_e^* has been taken to be less than m_h^*, so that μ_n is greater than μ_p. Why are the effective masses different in the two cases?

An electron in a semiconductor experiences various forces from the crystal lattice. The effect of these forces are taken into account by assigning an effective mass of the electron (m_e^*) in place of its true mass (m_o). Replacing m_o by m_e^*, we are able to ignore

these internal forces while evaluating the effect of the external force on the electron. Since the internal forces acting on the electron are different in the various crystal directions, the effective mass is also dependent on direction. To obtain an overall effective mass, a suitable average of the direction dependent effective masses must be taken.

From eqn.(1.10), we see that $g(E) \propto \left(m_e^*\right)^{3/2}$. N_C is also proportional to $\left(m_e^*\right)^{3/2}$. Therefore the effective mass to calculate effective density of states is taken as $\left(m_e^*\right)^{3/2} = \left(m_a^{3/2} m_b^{3/2} m_c^{3/2}\right)^{1/3}$, where m_a, m_b and m_c are the effective masses in different directions. On the other hand, from eqn. (1.66), we see that $\mu_n \propto \frac{1}{m_e^*}$. Therefore the effective mass to calculate mobility is taken as $\frac{1}{m_e^*} = \frac{1}{3}\left(\frac{1}{m_a} + \frac{1}{m_b} + \frac{1}{m_c}\right)$. It is very clear that the effective mass for calculating g(E) can be widely different from that for calculating μ_n. By similar argument, it can be shown that the effective mass for holes (m_h^*) used for calculating g(E) is different from that for calculating μ_p. It so happens that in silicon, $m_e^* > m_h^*$ for calculating N_C and N_V, while $m_e^* < m_h^*$ while calculating μ_n and μ_p.

1.5.2 Effect of temperature and doping on carrier mobility

From our discussion in the previous section, we observe that the drift velocity of the carriers stabilizes to an average value (instead of increasing continuously) proportional to the applied electric field, as the carriers lose their energy gained from the field by collision (scattering). The two basic scattering mechanisms that affect the carrier mobility are **lattice scattering and impurity scattering**. In lattice scattering, the carriers

moving through the crystal are scattered by the thermal vibration of the lattice atoms. With an increase in temperature, the thermal vibration becomes greater, and hence the frequency of such scattering events goes up. Thus lattice scattering effect dominates at higher temperature and the carrier mobility reduces as the sample is heated. Empirically, the mobility due to lattice scattering (μ_l) is expressed as a function of temperature is given by the simple relationship

$$\mu_l \propto T^{-3/2} \text{……..(1.68)}$$

On the other hand, impurity scattering is the dominant factor at low temperature. When a charge carrier travels past an ionized impurity, its path gets deflected due to Coulomb force interaction, i.e. due to the electrostatic force between the carrier and an ionized impurity. The probability of impurity scattering depends on the total concentration of ionized impurity and therefore the carrier mobility is reduced for a highly doped semiconductor. However, unlike lattice scattering, at higher temperature, the effect of impurity scattering becomes insignificant. At higher temperature, carriers move faster and therefore spend less time in the vicinity of an ionized impurity. Thus they are less effectively scattered by the ionized dopants. The dependence of carrier mobility due to impurity scattering (μ_i) on the temperature as well as doping concentration (N) is expressed as

$$\mu_i \propto T^{3/2}/N \text{……..(1.69)}$$

In general, the carrier mobility in a semiconductor can be expressed as

$$\frac{1}{\mu} = \frac{1}{\mu_l} + \frac{1}{\mu_i} \text{ ……..(1.70)}$$

Electron mobility in silicon for various doping concentrations of the sample is plotted as a function of temperature in Fig. 1.19. It can be seen that for lightly doped samples, mobility decreases with an increase in temperature, clearly showing the dominance of lattice scattering. However for heavily doped samples, mobility is low at low temperatures, where the impurity scattering dominates. It increases with temperature, reaching a peak, till finally at higher temperature lattice scattering takes over and mobility decreases again. Also, at any temperature, mobility is lower for samples with higher doping concentration. The variation of electron and hole mobility in silicon as a function of the doping concentration is shown in Fig. 1.20.

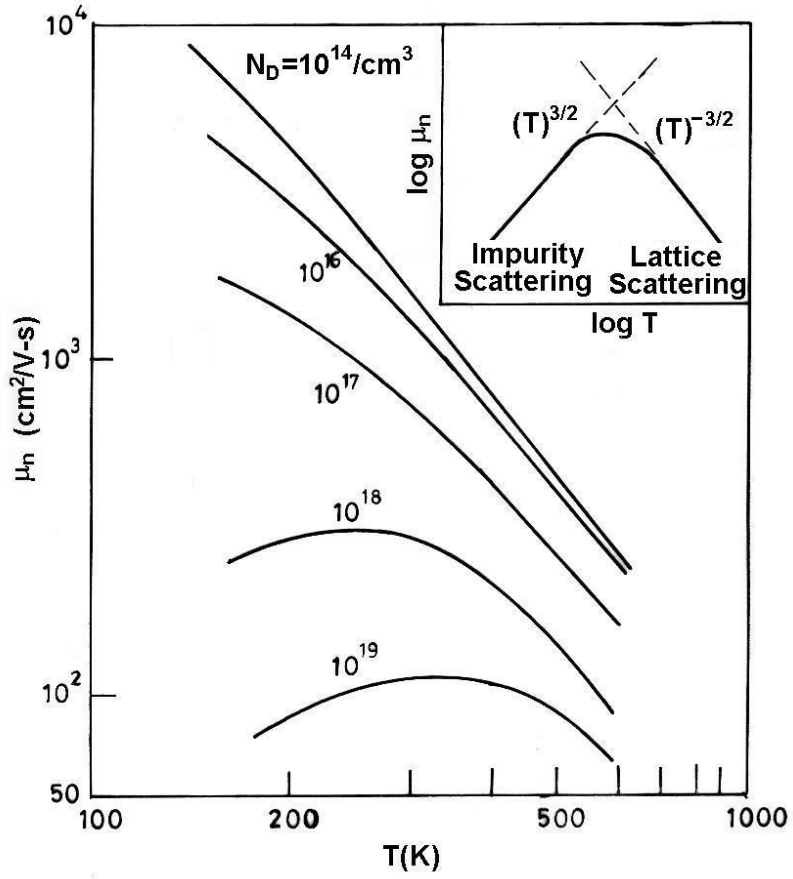


Fig.1.19 Variation of electron mobility in silicon as functions of doping concentration and temperature[2]

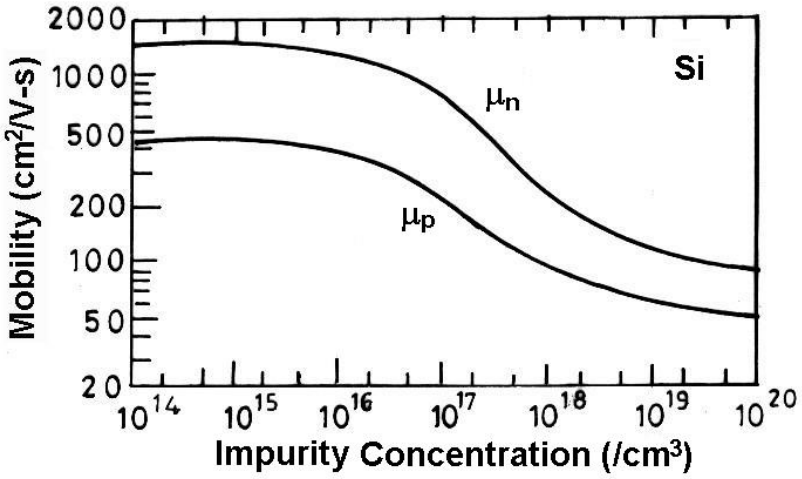


Fig. 1.20 Electron and hole mobilities in silicon at room temperature as function of impurity concentration[2]

1.5.3 Effect of high electric field on mobility

In Section 1.5.1, we have shown that the average drift velocity is proportional to the applied electric field. The proportionality constant is termed mobility, and it is independent of the applied field. However, this is true only when the applied field is small, so that $v_d < v_{th}$. At higher electric fields, when $v_d \approx v_{th}$, the drift velocity shows a sublinear dependence of the electric field. This represents the situation when the additional energy imparted by the field is transferred to the lattice rather than increasing the carrier velocity. So, the velocity approaches a saturation velocity (v_{sat}) called the **scattering limited drift velocity**. In other words the mobility is no longer a constant and its value decreases at high fields. The variation of v_d with electric field for silicon is shown in Fig. 1.21.

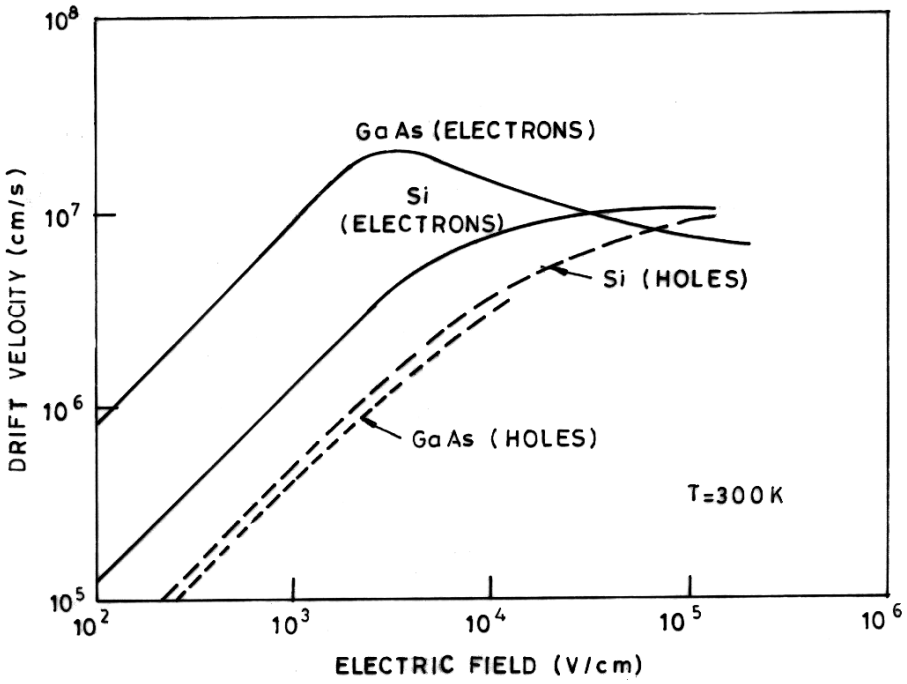


Fig.1.21 Variation of carrier drift velocity in Si and GaAs as a function of electric field[3]

For semiconductors like GaAs and InP the situation is more complex. There the carriers are transferred to a satellite valley with higher effective mass when moderately high electric field is applied and consequently v_d decreases. Thus for low electric field, v_d increases with E; it then decreases with an increase in the electric field till finally it reaches the saturation value of scattering limited velocity. This behaviour is also shown in Fig. 1.21.